

Exercise Sheet: Networking Fundamentals

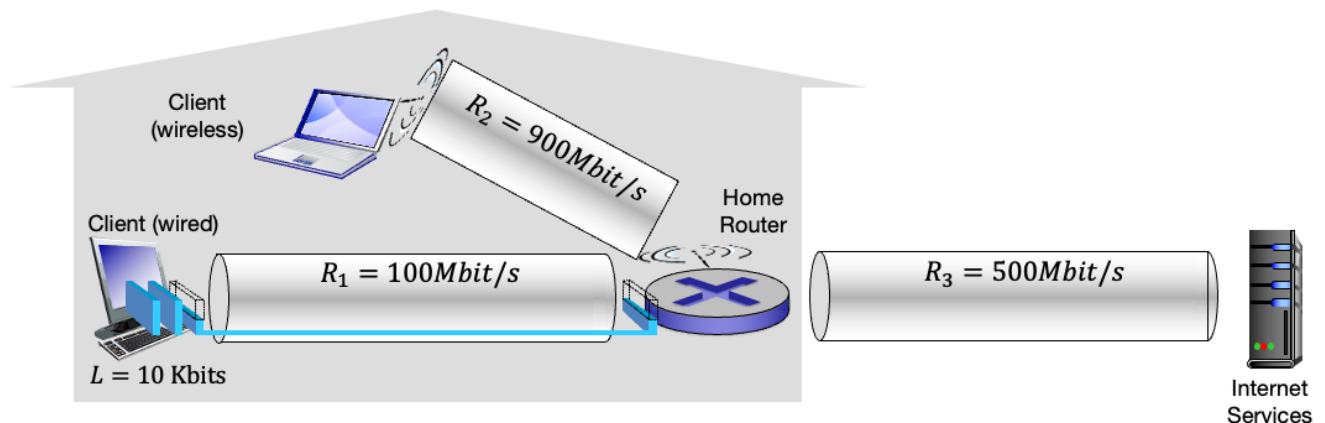
Exercise 1

Check the single correct answer in the following questions:

- 1.1: Describes the term “network edge” in your own words.
- 1.2: Describes the term “access network” in your own words.
- 1.3: Describes the term “network core” in your own words.
- 1.4: Which layer is the upper most in the current Internet protocol stack?
- 1.5: Describe the differences between packet and circuit switching.
- 1.6: How are ISPs interconnected?
- 1.7: How does packet loss occur?
- 1.8: Describe Cerf and Kahn’s internetworking principles and how they influence today’s Internet in your own words.

Exercise 2

Assume the network topology as shown below.



- 2.1: What are the bottlenecks in this network for the connection between (a) the wired client and the Internet Services, (b) the wireless client and the Internet Services, (c) the wired and the wireless client?
- 2.2: What is the packet transmission delay for connection R_1 .
- 2.3: What is the nodal delay for the home router, assuming the physical link has a length of 10m and the queueing delay is negligibly small.

2.4: How do the bottlenecks for (a), (b), and (c) in 2.1 change if the home router is replaced and R_1 has now a bandwidth of 1000Mbit/s?

Exercise 3

3.1: For what does a network protocol specify rules?

3.2: Why are network protocols organized in layers?

3.3: Explain the term “Encapsulation” in a few sentences.

Exercise 4

4.1: Explain the term “Forwarding” in a few sentences.

4.2: Explain the term “Routing” in a few sentences.

Exercise 5

5.1: Explain the term “Packet Sniffing” in a few sentences.

5.2: Explain the term “IP Spoofing” in a few sentences.

5.3: Explain the term “Denial of Service” in a few sentences.

5.4: Which categories of the STRIDE model from the Cybersecurity Fundamentals lecture does each of the three terms above correspond?